Symmetries in Physics — Exercise Sheet 2

Structure of groups, quotients, and lattices

Variant: Questions only

Exercise 1 (i) Show that every subgroup $H \leq G$ of index 2 is normal. (ii) Deduce that in any finite group $G \leq \mathrm{O}(n)$ containing an element of determinant -1 (for $n = 2$, a reflection), the subgroup of determinant-1 elements — the *rotation subgroup* — is normal of index 2.

Exercise 2 Compute the conjugacy classes of (i) D_4 and (ii) S_4, and verify the class equation $|G| = \sum_i |C_i|$ in both cases.

Exercise 3 Use the first isomorphism theorem to identify the following quotients: (i) S_n/A_n for $n \geq 2$; (ii) $\mathrm{GL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{R})$; (iii) $\mathbb{R}/2\pi\mathbb{Z}$.

Exercise 4 Show that $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/mn\mathbb{Z}$ if and only if $\gcd(m,n) = 1$.

Exercise 5 Find the centre $Z(D_4)$, and show that the quotient $D_4/Z(D_4)$ is isomorphic to the Klein four-group.

Exercise 6 (crystallographic restriction) Let $L \subset \mathbb{R}^2$ be a lattice (the integer span of two linearly independent vectors) and let R be a rotation with $RL = L$. Show that the order of R is $1, 2, 3, 4$ or 6 — no periodic crystal lattice has fivefold symmetry.

Exercise 7 (semidirect products) Let $N \trianglelefteq G$ and $H \leq G$ satisfy $N \cap H = \{e\}$ and $NH = G$. (i) Show every $g \in G$ is uniquely $g = nh$ ($n \in N$, $h \in H$), and that the product reads

$$(n_1 h_1)(n_2 h_2) = \left(n_1\, h_1 n_2 h_1^{-1}\right)(h_1 h_2)$$

— the *(internal) semidirect product* $N \rtimes H$, with H acting on N by conjugation. (ii) Exhibit $D_n \cong C_n \rtimes C_2$ and identify the action of C_2 on C_n.

Exercise 8 (composition series) A *composition series* of a finite group G (Lecture 2, where the chain is written from G downwards; here the same chain is read from $\{e\}$ upwards) is a chain $\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$ in which each quotient G_{i+1}/G_i is simple. (i) Exhibit a composition series for D_4 and list its factors. (ii) Find a *second*, genuinely different series for D_4 and check it has the same factor multiset — the Jordan–Hölder theorem in miniature.